

The NFL Project

Predictive Modeling for Players's Future Performance

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Introduction

For those of us football fans, it is a source of great fun and discussion to debate how long particular players of note will “stay good” for. A gentleman like Patrick Mahomes is right at the top of his game right now. But for how many more years will he be the incredible player he is right now? Five more years? Ten more years? Barring injuries, his career ought to follow some natural progression as he ages. We know that football players tend to slowly fade off in their amazing skills as they approach their 40s, many retiring in their late 30s, but beyond that very vague understanding, it often seems to me that any more detailed predictions are based simply on speculation and conjecture. Thus I set out on this project to see if it's possible to use data mining techniques on all the historical NFL player info available to try and make a predictive model that can better address the question of how long a particular player will “stay good” for.

My goal was to be able to pick a particular modern quarterback in the middle of their career, Josh Allen of the Buffalo Bills lets say, and use the predictive model to guesstimate what the rest of his career might look like.

Abstract

This project developed a predictive model to estimate the career trajectories of NFL players. Using historical NFL player data I was able to scrape from two different truth-worthy online databases, I applied data mining techniques to build a comprehensive predictive modeling system from scratch. I am profoundly excited (and quite surprised) by the accuracy of the machine! I was able to test its veracity by taking past NFL players long retired, plugging 1/2 or 1/3rd of their career into the model, having it predict the rest of their career, and finally comparing that output to what really happened. These tests showed that over 70.2% of the time, the model was able to predict the future of that players career to within $\pm 10\%$ of what really happened. This imply that the model is a robust and reasonable way of actually assessing the future career trajectories of current NFL players!

In addition to career forecasts, undertaking this project has provided insights into how the game of football has evolved over the decades, including shifts in passing and rushing trends.

Literature Survey — Related Works

While there is big business in sports news and radio broadcasting about discussing, arguing, and speculating about various players's expected performances, there seems to be a surprising dearth of actual scientific research on the topic. I would gently conjecture that based on the high dollar value of investment teams make on various players and the kinds of multi-year contracts they sign, I can only imagine that there's actually *extensive* research on the topic! But those techniques and formulas and predictive models are all proprietary info, locked behind the closed doors of the scouting offices and general managers's offices of the various teams. Very little of that information seems publicly available.

Investigating the topic, I was, however, able to find the following scientific papers on this subject.

- In 2021 a man named Carlos Fonseca did a detailed data-science-focused investigation [1] into a much more specific question: "Can Tom Brady continue to play football until he is 50 years old?" He used a lot of statistics, curves of best fit, ... he did good work! He ultimately had to wrestle with one of the most interesting and difficult aspects of this whole topic (and one I'll face extensively in my own research!) — the question of how does one quantitatively analyze what it means for a player to be "still good". There are many various statistics one can use to measure a player's performance year by year, and no one of them tells the full story. He had to, as I will as well, come up with a way of amalgamating all those various figures into one single number. This is the part of his research that most overlaps with my own, although I currently dream of doing that amalgamation in quite a different fashion.

His research, while interesting, was ultimately all about one single player and won't help me with my project too too much. My work will seemingly require a lot of ingenuity and off-road creativity.

- Another area of research focus that one finds in the literature is that of trying to predict how well a graduating college football star will do upon entering the professional leagues. Barbatsis, Rhee, and Zhang's 2024 article [2] is a fine example of this. They once again construct an elaborate data science model for predictive purpose. Their model looks back only at the last few years, but fairly intensely scours the data. It looks at how various recent college football stars did while in college, how they performed numerically at the scouting combines, and then how well (or terribly!) they did upon reaching the NFL.

I don't believe I could use their particular techniques to address my question, as their examination is really mostly trying to figure out which players will thrive or flop in their first year or two in the league. I'm trying to examine the whole future of a career; quite a different question... But perhaps the most relevant takeaway from reading their work is that each one of us researchers on this topic seems to be constructing our own predictive model from scratch, rather than using some established, known, trusted model for the job. This gives me hope that I'm not going in the wrong direction.

- Lastly, perhaps worthy of mention is one more article I found on the topic by a Mr. Steven Pintado [3] who is also using data mining techniques to examine NFL football players's performances, but in his case he's looking at how to estimate an upcoming season's NFL *Fantasy* Football performance based on previous years's. His work is interesting, but this is

quite a different question, and I believe it to be a little too far afield to be much helpful in the work I'm doing.

Proposed Work:

- (1) There are no publicly available-for-download data sets on NFL statistics, so I had to strip them myself from a series of websites. This was an interesting and difficult endeavor and pushed my understanding of recent things learned in CSPB 3308 Software Development Methods (bash scripting, regular expressions, etc) to the limit.
- (2) The further back one looks, historically, the more shaky and (strange?) the data looks? I imagine it's because the game of football has changed so vastly that it's hard to even compare modern carefully-recorded player statistics with the kinds of shorthand notes people were taking about the game in 1940. But were I to take the data at face value, it looks like only a few players were even playing in the 1940's? And they ran for very few yards and threw for even less? For this project, I was going to have to do some historical deep-diving to try and make sense of the earliest data. As it currently sat, it was slightly confusing.
- (3) Next up, I had to integrate at least two different data sources of NFL player data to form one large data cube. I was sure there would be some discrepancies, and I'd have to log in some work to get to the bottom of them.
- (4) I was sure over the course of the 100+ years of NFL historical data there'd be some missing entries. I'd have to come up with ways of sensibly patching the holes.

- (5) Lastly, the game has changed enough that I'd have to come up with ways of addressing things like the fact that there are now far more games each year than there used to be! When examining how many passing yards a player got over their career, I'd have to account for the game itself changing over those same time spans. This introduced some complexity and required lots of creativity to solve.

Dataset:

Early work resulted in me finding two online resources that I could use to build my data set from.

- (1) I could get information directly from the NFL itself off their designated statistics website: <https://www.nfl.com/stats/player-stats/> [4]. This site is great, and it had a lot of the information I was seeking. It has all of the various player stats sorted by year, by player, by position (quarterback, runningback, etc), and by the type of information it is (passing info, fumbles info, tackles, etc). This was great! ... But would also require quite a lot of work to parse into a useful data cube for my project. A strong limitation of this website is that the info only goes back to 1970. I'd have to use the following, less good, resource to gather data from earlier and knit it together with the info from this one.
- (2) The next resource I found is from <https://www.footballdb.com/statistics/nfl/player-stats/>. This is a more gritty, less user-friendly website of pure info. It is less fun and flashy, but that was fine for my project, and it seemed (for the most part?) to agree with the data from the former site. Most importantly, this website's information dates back as early as 1940, so I should be able to knit the two

together to get a continuous stream of player data from 1940 to 2025 to work from.

Across those two websites, I should be able to construct a data cube with somewhere around 20 to 30 dimensions and containing a few hundred thousand data points. This is exciting!

Evaluation Methods:

The whole point of this project was to answer the question “is it possible” to predict future player performance. It was certainly possible that it’s not possible! That uncertainty might have been the most fun and exciting part of this project; it was an actual open question!

The classical advice, found in the course lectures, would have been to leave approximately 20% of the data out of the training of the predictive model and save that aside as test data. The trouble with using this method here is that there are only hundreds to thousands of NFL players, historically, not millions. I needed every single player I could find to train the machine on! I ended up deciding to use sampling-with-replacement as a wiser plan. Thus for each player I wanted to test the machine’s efficacy upon, I’d simply remove them from the training data at the time of the test, perform the test, and then insert them back in.

Let’s say one of the chosen players in the test sample turned out to be Brett Favre, who famously played for the Green Bay Packers. Let’s say we snapshotted his career at the 10 year mark and asked our predictive model what the rest of his career looked like? I wanted it to spit out things like how many more touchdown passes did he throw? How many more

interceptions? When did he retire? And lastly (and most importantly for our research question), I wanted it to estimate “for how many more years did he play well for”? ... It is that last question that I was going to be evaluating my model based on. In the best of worlds, I wanted it to be accurate to within ± 2 years or so, or, to state it differently, accurate to within an error margin of $\pm 20\%$ or so.

I was profoundly excited to see how well I could get a predictive model to predict player performance!

Tools:

Originally I considered using higher-dimensional K-Means Clustering to sort the players into clusters of like players. This seemed powerful and effective, but then it occurred to me that it might generate very boring results. Suppose all quarterbacks who are fairly good, playing in 2025, happen to cluster together. We might get the same exact prediction for all of their future careers!

I decided to invent my own style of “clustering”, so to speak, by gathering a list of the closest 30 players and making a kind of “cluster on the fly”. This cluster, I could then find mean-data from and make my predictions.

For stripping the info off the websites, I would use bash scripting, regular expressions.

For cleaning the data and building my model, I would be using lots and lots of Python coding.

Gathering the Data:

- The first tremendous task on this project for me was stripping all the raw data off the two various NFL statistics websites. One site, FootballDB.com, has its information nicely capturable on easy-to-strip

pages. But this is partially due to the fact that it only stores info on the top handful of players each year. Starting Quarterbacks, lets say. The other website, NFL.com, has a record of ALL of the player info. Every time a wide receiver threw a surprise pass as a trick play, etc etc. This was invaluable info when it comes to gathering slightly more off the beaten path info like historical data on the number of yards per season various quarterbacks *ran* for in the 1970s, etc. And I certainly needed to tap into that significantly more vast ocean of data to find tackle and forced fumble data on all the defensive players — thousands of them per season!

The tremendous road block I encountered was that my primary source of info, NFL.com, presents all the info on interactive charts that have a “next” button at the bottom that you have to click to scroll further down the chart. When trying to compose Python code to automatically strip all this info, this is a *nightmare*, as none of the classical HTML stripping tools help. I had to go into a nearly four-week journey into learning tools like Selenium and JSON API endpoints to almost build a kind of “bot” smart enough to wander the NFL.com website all by itself and gather data and click the next button. Additionally, half the time you click the next button either ads pop up or a menu asking if you accept cookies pops up, and the Python bot has to close those before proceeding. It was brutal work trying to get it to run, and I was pretty convinced at various points that I would be unable to get it done.

But miraculously, I pulled it off! Part of the process required leaving my computer running, without sleeping or screen saver, for over 4 days gathering all the NFL player data! It took this long, as the

“next” button has a glitch where if you click it too quickly, it does not load the next page at all but stays on the current. It takes 15 seconds of waiting each time to be able to continue scrolling down the page.

At the end of those 4 long days, I had gathered all the data I would need for the project.

Data Integration:

I now had all the raw data for the project, but it was currently contained in 12 different .csv files, with vastly different column titles. The info from 2024 contained many different stats that were not recorded in the 1980’s, and similarly the 1980’s had much better record keeping than the 1960’s and 40’s. Everywhere you looked, the .csv files were in conflict and unintegrable. Another week or two’s work went by as I composed no less than 38 different python programs to help me take all these varying statistics and build a massive data cube. (These programs, if interested, are found in my GitHub repo). I had to rearrange, chop, snip, and sort them lists in all kinds of different ways to get them properly organized enough to be combined. Some of these were techniques covered in the lecture and in the book. Some I had to invent myself for unique circumstances that arose.

Data Cleaning:

Early into the work, I began to see a strange and uncomfortable pattern. There was something wrong with the year 1990. I’ve since researched it extensively and have found nothing historical to explain it... But in the year 1990 many data entries are randomly all 0s. Perhaps even more bizarrely, for some specific statistics, 90+% of players have a false

0 and a few stragglers still have normal-looking data for that year!

I decided, since it was only a single corrupted year, to try and “patch over the hole”, so to speak. I considered: (1) Taking each player and making 1990 either echo the stats of 1989 or 1991, (2) Simply deleting 1990 from consideration all together... I finally decided to be brave and do something more difficult. (3) I decided to try and extrapolate appropriate data for those corrupted values in 1990 by taking the uncorrupted data values for each player that year, scanning across all available non-1990 data in the whole database for the most-similar entry, and filling in the corrupted data values with the equivalents from that similar entry. This took some clever coding.

It’s important to mention that as one drifts back up in the data, many statistics begin to vanish off the charts. Those statistics just weren’t measured a long a time ago! I briefly considered perhaps extrapolating those values (?), but that seemed like far too much of me generating “imaginary” data. I wanted my project to be as substantial-by-the-facts as I could get it to be.

Major Choices:

Along the way soon came many difficult and interesting choices. Many of these I spend chunks of days thinking about.

- A first example would be... what to do with playoff data? Every year some players make the playoffs, many don’t. Some only play one playoff game and lose. Others end up playing four such games and winning the Super Bowl. What should we do with this data? I considered possibilities ranging across: (1) Simply reward playoff players by adding their

playoff statistics on *top* of their regular season statistics. They did throw and catch the ball more, after all. ... I eventually decided against this, however, as it seemed to pull too hard for the players who either got lucky or whose team *around them* played well. (2) I considered having playoff statistics be kept and stored separately from the regular years all together. It did seem relevant to keep track of how a player played in those big playoff moments that really mattered. Some players are famous specifically for being clutch in those key big games. ... But this would be difficult to work with, as NFL.com's database (with far more players) doesn’t show playoff info while FootballDB’s does. And FootballDB is the site with the incomplete information only on some players. It seemed random to reward players who I *had* info on over ones who I simply didn’t have the info on. (3) I finally decided that in pursuit of objectiveness and fairness (and thus better model predictive power) I ought to leave playoff info completely out of this model. It broke my heart just a little, as I still think that data does matter in the discussion, but I could find no other way to proceed.

- A second massive choice was the discussion of *how* precisely I was going to measure the similarity between two players. Ultimately I wanted to somehow measure “how good” a player played that year, perhaps in some kind of an A+, B- kind of grading system. But how was I going to convert the rows of the data cube into such a grade scheme? The problems were many-fold. I considered making some sort of a formula adding together the various statistics with weight-constants attached to them; points awarded for touchdowns and yards gained,

points subtracted for interceptions and fumbles lost... But this was going to expose the key differences in how the league has changed over the years. A 2000-yard passing season would have been record shattering in the 1950's. Now that would classify as quite sub-par. Maybe I should normalize the values *within each year* (?), comparing like players to like players? But even that poses the problem of how to compare a throw-heavy quarterback vs a run-heavy one.

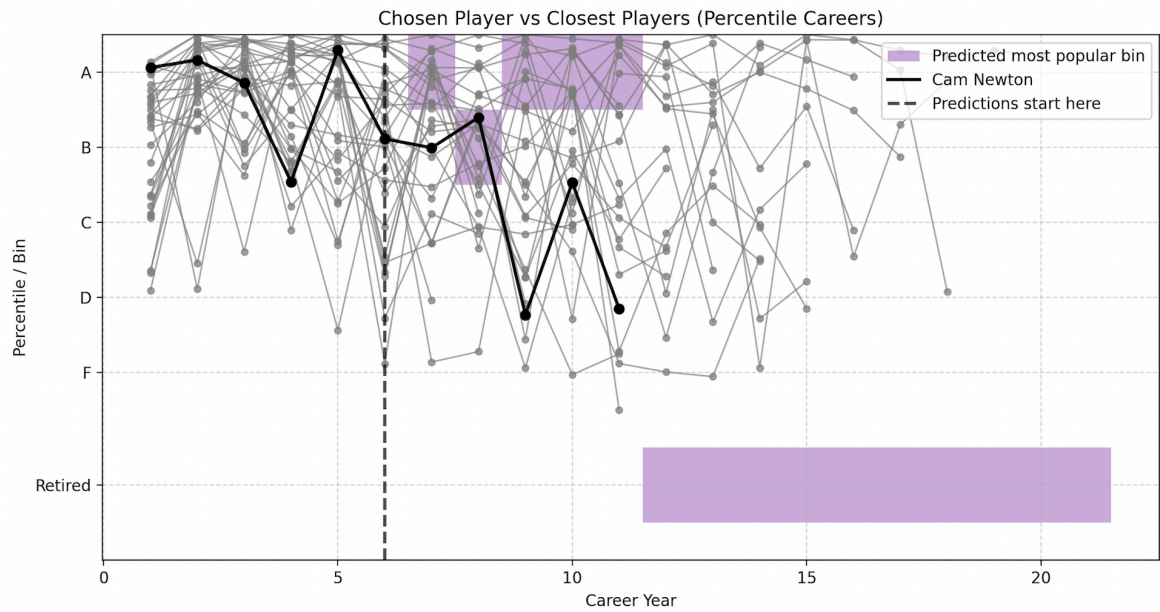
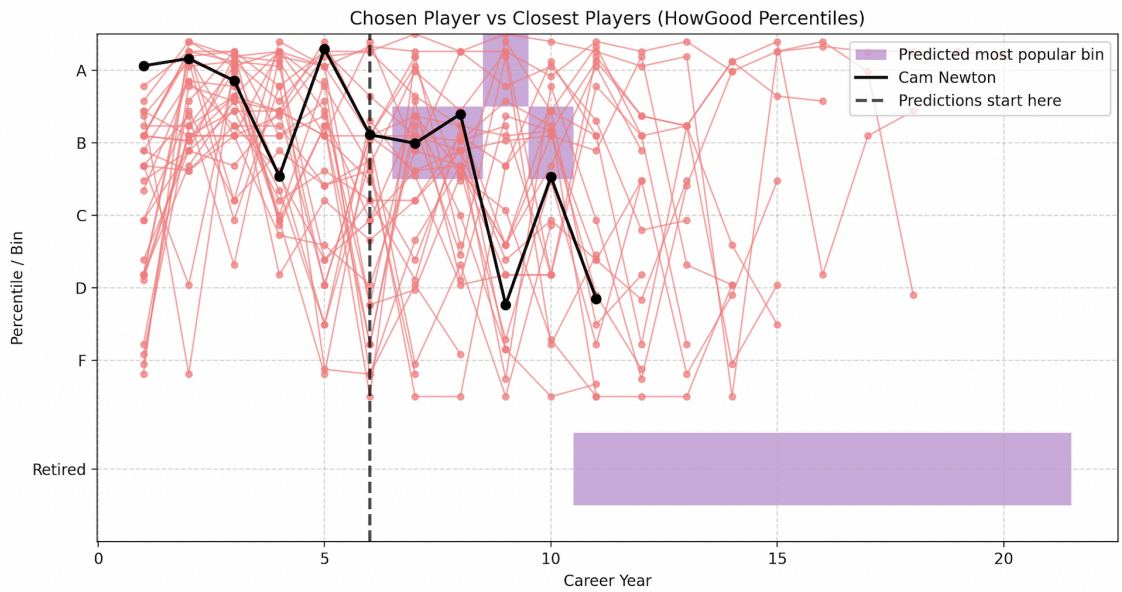
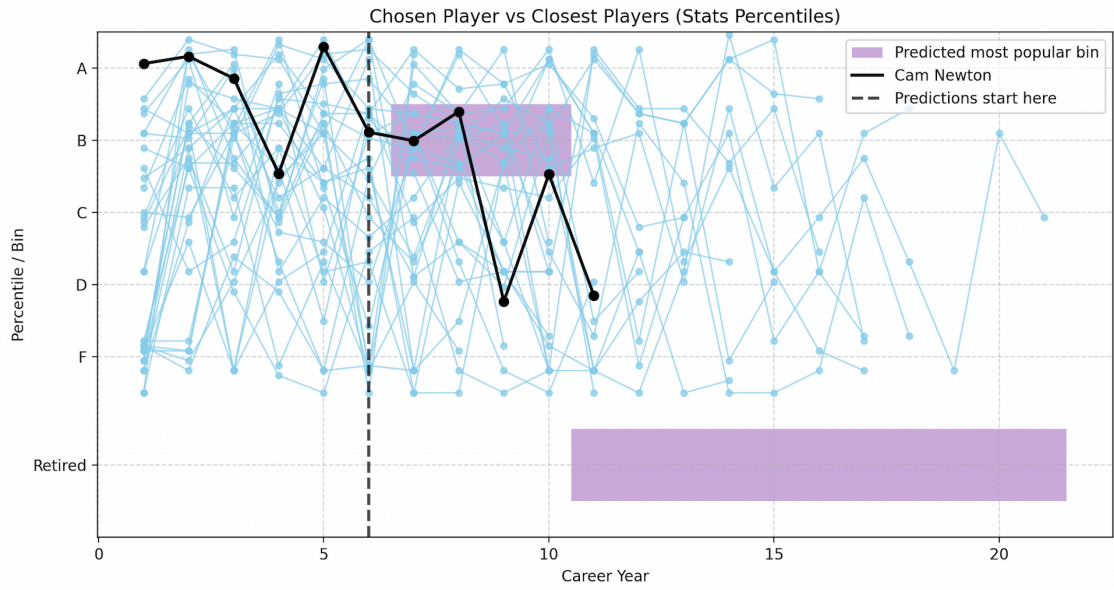
The discussion of how to compare the various players got so over my head, I decided to actually pursue three different possible solutions and perform a small study to see which technique gives the best results! The three avenues I chose to pursue were (1) Directly comparing player statistics. Find the players who literally did the most similar things during their time in the league, by the numbers. (2) Use an amalgamated formula to calculate a number for "how good" a player was in any given year and then compare those. (3) Or instead use the data I had to *rank* each player numerically in the order of how well they played. Rather than looking at the actual "how good" raw score, we'd simply give each player their percentile ranking for the season, #1 in the league down to last place, and compare those. These three different solutions ranged from the most tangible (the stats method) down to the most abstract (the percentiles method).

It made the project more involved pursuing all three, but I was very curious to find out which was the best method.

Key Results:

Multiple weeks of Python coding later, I finally constructed my machine and was finally able to boot it up and try it out! Let me give you a few examples of what it looks like.

Let's first test it out on the old Panther's running-heavy quarterback Cam Newton. Cam Newton played in the NFL for 11 seasons, and I decided in this experiment to give my predictive model the first 6 seasons of his career to base its predictions on. On the next page are the three data printouts that my machine gives for him.



In reading one of these charts, to the left of the black vertical dotted line is the given data we offered the machine and to the right of it is its predictions. The black line is Cam Newton's actual career, while all the other lines (blue, red, or gray) represent those of the 30-closest players in the data base. The blue one is one for closest stats, red for closest "how good" raw score, gray for the closest percentile year by year.

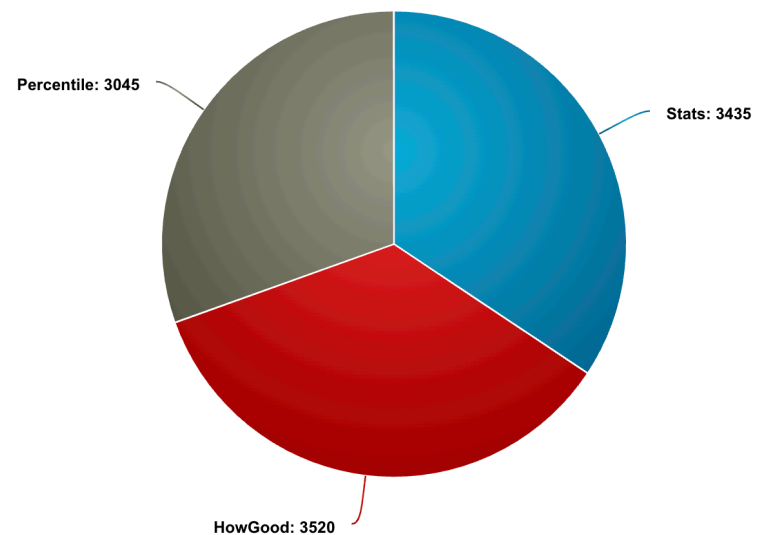
In the first iteration of my machine, I attempted to use the *mean* score for these 30 players, but it sadly gave an almost unusable prediction every time! The players who retired early completely obliterated the average who retired early completely obliterated the average down to near zero every time! This was an absolute shock to me (jaw dropped), as I thought I'd be able to use the lowered score from the mean to *help* us predict when a player would retire. At the very last minute I had to completely re-think my entire design.

I decided to use another technique discussed in class and bin the various scores into their various letter grades and use the *mode* instead of the *mean*. To let my machine make its predictions. You can see in the above graphs that it is taking the 30 closest players, plotting their careers, and taking the mode of each year's binning. In short, the purple bands now represent both the mode and the model's prediction.

And you can see it does a pretty good job! This is just one randomly chosen player, but I'd highly recommend taking my little python code out for a spin. It really works! It tends to predict player talent for the next 3 or 4 years very accurately, becoming slowly less effective the further out you want it to look (that makes sense). It tends to get both the estimated future performance accurate to within one letter grade and the estimated year of retirement

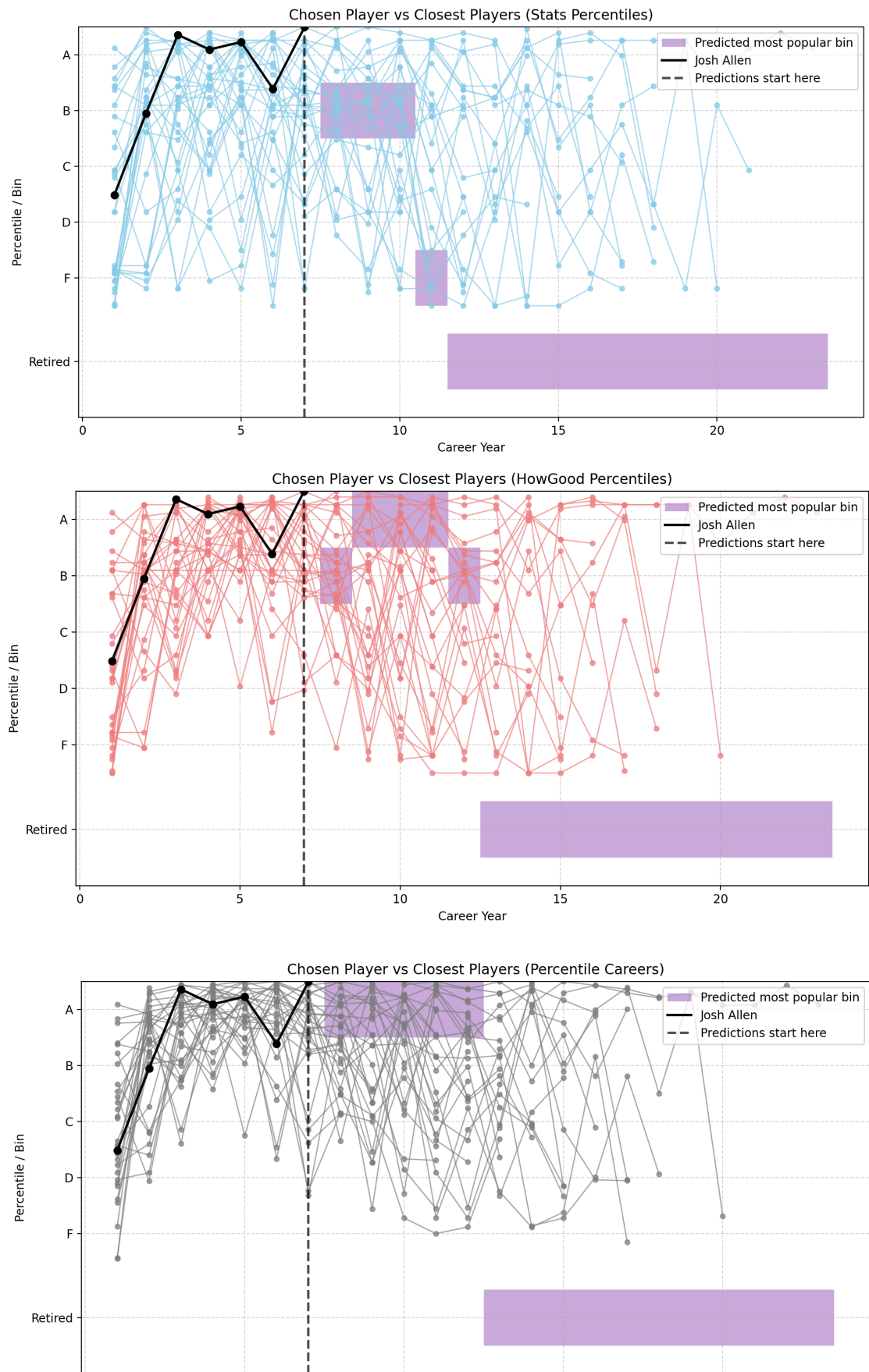
accurate usually to within ± 1 , sometimes ± 2 . I ran some extensive tests and found that 70.2% of the time it gave a prediction within $\pm 20\%$ of the truth. After so much time wondering how well (or how terribly!) the machine would perform, I was fairly thrilled with the results.

I ran a whole extra set of automated simulations to try and calculate which one of the three methods of grouping like players is best — stats, how good raw score, or percentile... and the results were quite unexpected! After running 10,000 simulations and seeing which of the three was closest each time, the results were a near 1/3rd, 1/3rd, 1/3rd distribution!



Stats and HowGood barely tied for first. The above was not at *all* how I thought that experiment would go! I'm still not quite sure how to process or incorporate the above into my research, as it doesn't really answer the question of "which one is the best technique to use". For the moment, I've just been using all 3 techniques to give three slightly different predictions. I like to pretend they're the opinions of three of ESPN's talking heads, arguing.

We're now finally in a position to answer the original question that started all of this. For how long will Josh Allen, quarterback of the Buffalo Bills, continue to play well for? Here's what my data mining model has to say about that:



You can see that all three models agree that Josh Allen will play for about 4 to 5 more seasons. One model suggests he'll play pretty good (B-grade) for a few seasons with one last not so good one. The other two models suggest he'll play excellently for 5 more seasons right until the day he retires.

After so much work to build this machine, it gives me no end of joy to run it and see what predictions it makes! It'll be very fun to look back on this project a few years from now and see how accurate these predictions turned out to be!

Limitations:

There are a few key things that my machine does *not* do well, and for full fairness and honest disclosure, I thought it would be worth mentioning them.

- From the moment I first started to compute HowGood raw scores for players, it was very hard to compare significantly different players. How does one properly compare a running quarterback like Lamar Jackson to a great but stationary one like Payton Manning? It just never quite works. All my calculations slightly inflate the value of doing *more* different things at a B-grade level, rather than doing one specific thing at an A++-grade level. So far, I've found no way to counteract this in the data mining.
- On the defensive side of the ball especially, I've simply short changed some players. Some of the big lads up front on the defensive line (defensive tackles, let's say) are tossed in the same database with flashy cornerbacks and linebackers and are being graded on their number of interceptions and tackles and such. But this is truly not fair to them. Their job is to PUSH as hard as they can and stop

the run. It seems mightily unfair to me that an amazing defensive tackle will show up in the data as approximately equivalent to a mediocre cornerback. In short, the defensive data is best suited for cornerbacks, safeties, and linebackers, and not for the big lads up front.

- And lastly, and most meaningfully, there is something critical here that I feel like needs be said. A human being is more than the collection of their statistics. There's a player right now in the NFL named Joe Thuney. He's an offensive lineman. While his statistics wouldn't much impress anyone, his teammates say he's an absolute *monster* of a personality. He's so passionate, so feisty, so believes in what he's doing... He inspires everyone around him to hit the gym harder, to truly try their hearts out. He's a powerful spiritual leader and a force of a personality on the field. A complete field general. Nothing in all my data cubes could ever account for the human element of what makes Joe Thuney magnificent. Being more than our data is one of the things that makes us human.

Applications:

While I'm not sure I'll be getting any calls from ESPN any time soon, desperate for my predictive model to help them with their broadcasts, I think there are two very real applications this little project has.

- (1) My friends and I will get some glorious laughs and good times in the month to come looking up various players and predicting what the prediction will be! It'll lead to much cheer and merriment.
- (2) As a student about to embark upon a Masters in Data Science, building my own extensive data mining machine entirely from scratch has been the

most interesting and engaging school project I've done in a long time! I learned SO much while building this machine, and it brought almost every one of the concepts from throughout the course to life. Quite literally this course and this project is one of the main things that convinced me to pursue Data Science over Software Development. If I have a nice life & career out there someday, with a little imagination some fraction of that future could be considered "an application" of this very project.

Thank you for the opportunity to do this insane and over the top data mining project. It meant the world to me, has been truly inspiring and fun in my life for months now, and I'm very grateful to have gone on the journey.

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